Slice of pi

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NSU Sys.Pro

Harness

Harness

What it does?

Harness

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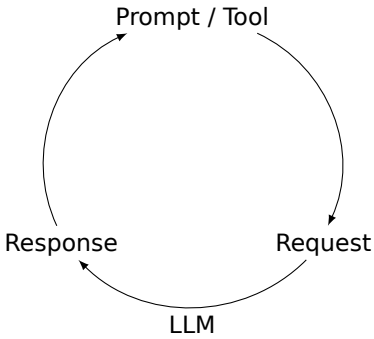
- LLM communication
- Skill access
- Tool execution
- Context management
- Session management
- Chat display

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- Agentic loop

Agentic loop



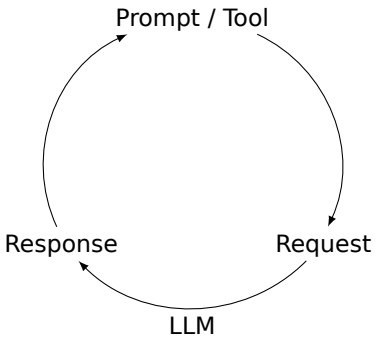
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Examples

Agentic loop



Harness

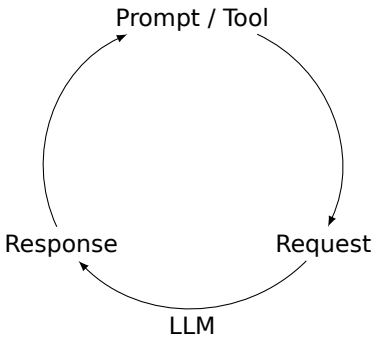
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Examples

- Claude code
- Codex
- Opencode
- **pi**

Agentic loop



Context

What is context?

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- Initial prompt
  - System prompt
  - Tool descriptions
  - Skill descriptions
  - MCP descriptions
- Messages
  - User prompts
  - LLM responses
  - Tool call results
  - LLM reasoning

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- Keep last 20k messages without reasoning
- Ask LLM to summarize everything else

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System prompt	2 742 ch	8 603 ch
Tools	4 · 2 956 ch	12 · 14 033 ch
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Tokens	1 395	5 077
Price	₽0.027	₽0.090

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- Claude code/Codex prompts are even bigger
- Increased initial prompt size is better for model providers
- Initial prompt is what survives compaction
- Bigger context on average costs more

pi agent harness

There are many agent harnesses but this one is yours

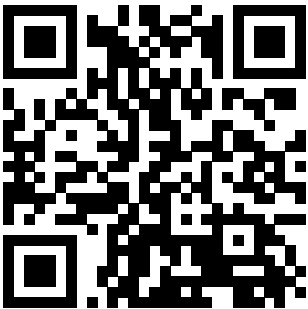
pi agent harness

There are many agent harnesses but this one is yours

- Minimal
  - Small system prompt
  - 4 tools by default
  - No MCP/sub-agents
- Extensible
  - Vibe-code extensions/skills/themes
  - Package marketplace
- Transparent
  - .jsonl session files
  - Good bundled docs
  - Extensions can access all internal info

https://pi.dev

My pi config



https://github.com/liontiger23/configs-pi

My pi config

My extensions

- `routerai-provider.ts`
  - Model selection UI
  - Shows real cost
  - Model provider selection UI
  - Sorting and filtering
  - Limits context size
- `turn-timer.ts`
  - Tracks current turn stats
  - Time, cost, inference speed
  - Reports summary when model turn ends
- `pin.ts`
  - Pin arbitrary text or skills to system prompt
  - Survives compaction
- `no-root-scan.ts`
  - Blocks searches in root of file system



https://github.com/liontiger23/configs-pi

My pi config

My skills

- `duckduckgo-search` - Web searching
  - search.sh + fetch.sh
  - curl + w3m internally
  - Uses GitHub CLI for GitHub access
- `pair` - Pair programming system prompt
  - Plan then implement
  - Do only what is asked
  - Pause on surprises
  - Don’t deep dive
  - Autonomy when explicitly asked
- `tutor` - Tutor system prompt
  - Does not give you the answer right away
  - Guides with questions
  - Reduces output size
  - Combines with `pair`
  - Original source



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Q&A